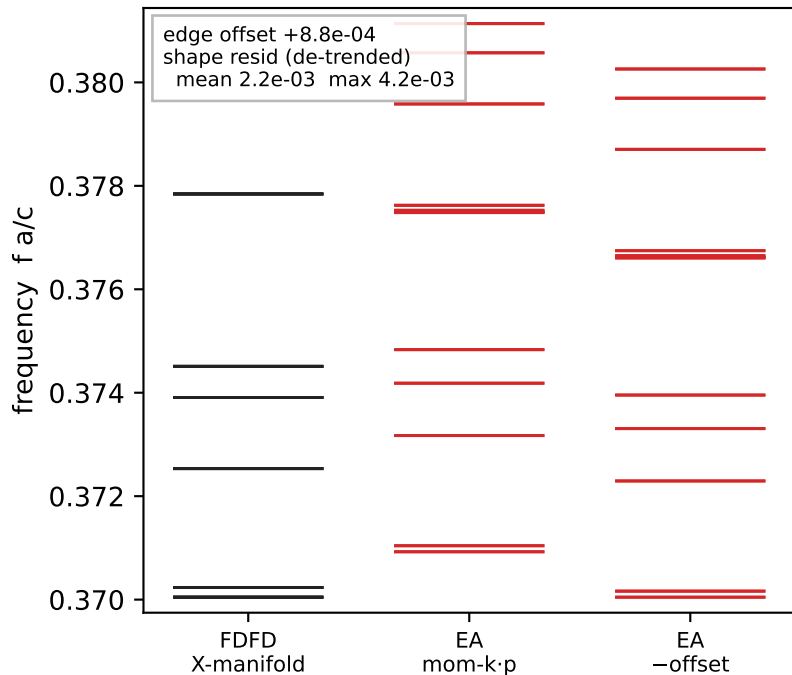
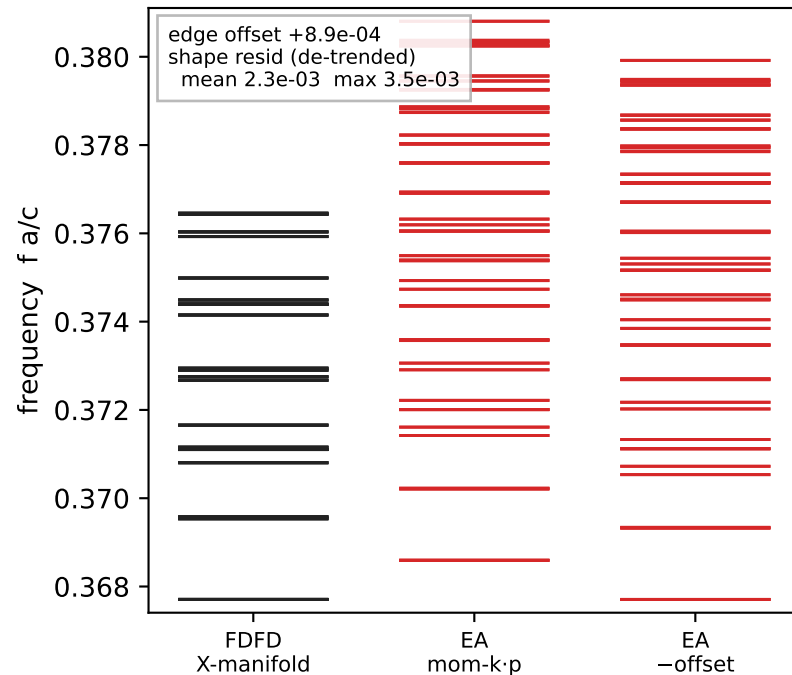


Momentum-space $k \cdot p$ EA reproduces the FDFD moiré manifold — square CML TM, band-1 gap edge

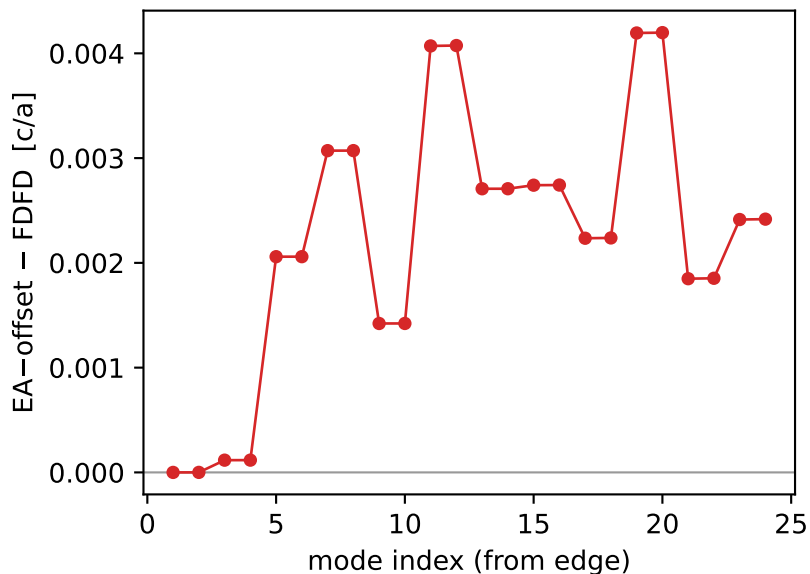
2deg ($m=57$) $\theta=2.010^\circ$, $\eta=0.0351$
FDFD-X manifold: 24 states (index-aligned)



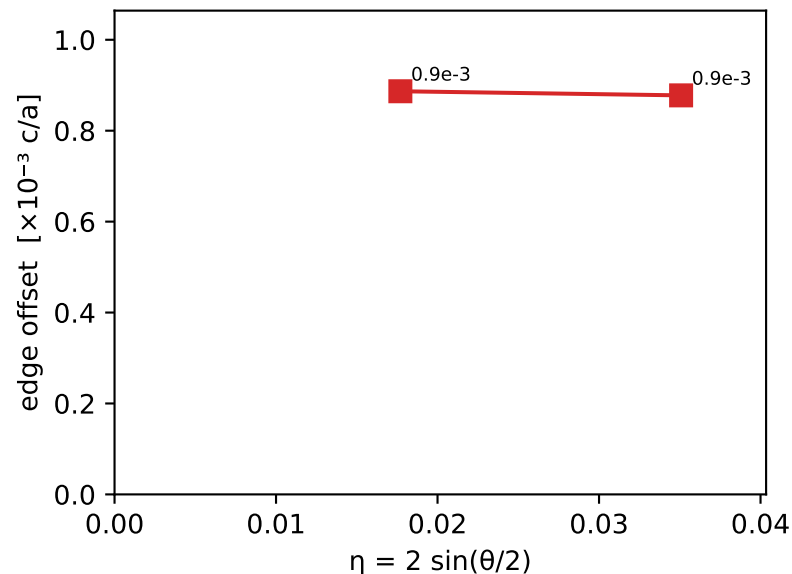
1deg ($m=113$) $\theta=1.014^\circ$, $\eta=0.0177$
FDFD-X manifold: 88 states (index-aligned)



2° miniband-shape residual
(rigid offset removed)



offset $\rightarrow 0$ as $\theta \rightarrow 0$
($\propto \eta^{-0.01}$; separable-approx residual)



Momentum-space $k \cdot p$ moiré model

$$H_{GG'}(k) = E_{\text{ref}}(k_0 + k + G) \delta_{GG'} + \tilde{V}(G - G')$$

E_{ref} : EXACT local band-1 dispersion (MPB, ref registry) – replaces the parabolic $k \cdot p$ that over-binds $\times 9$
 $\tilde{V} = \text{FFT}_s[\Lambda_1(X; s)]$ (phase-1 landscape)
plane-wave cutoff $|n| \leq 5$, 4 moiré lanes

Result: over-density GONE; $\times 4$ quadruplet structure + count reproduced;
residual = rigid offset ($\rightarrow 0$ as $\theta \rightarrow 0$) + $6.5e-4$ miniband shape at 2° .

Square CML, TM (E_z) | ASYM bilayer: layer1 $r=0.20$ + layer2 $r=0.10$, $\epsilon=8.9$, $\epsilon_{\text{bg}}=1.0$ | registry-common gap $[0.3225, 0.3661]$, band-1-at-X manifold isolated (band-2 headroom 0.12)
EA: momentum-space $k \cdot p$, $k_0=X$, E_{ref} from MPB res64 at ref registry $s^*=(0.230, 0.105)$, $\tilde{V}=\text{FFT}$ of reg128 $\Lambda_1(X; s)$, cutoff $|n| \leq 5$, moiré lanes $\{\Gamma, X1, X2, M\}$ of the 2×2 CML fold, pooled
FDFD: $L=\epsilon^{-1/2}(\sum g^{\{ab\}} D \dagger D) \epsilon^{-1/2}$, subpixel $N_{\text{sub}}=8$, CHOLMOD shift-invert, $Q_X=(\pi, 0)$; filtered to $w_X + w_{X'} > 0.5$ (X -star Fourier weight, empty-lattice-calibrated $k \cdot b$ map)
STRICT: sorted ladders, index-aligned from the edge, no Hungarian; the $\times 4$ clusters are valley($X \otimes X'$) $\times C4$, per the centered-cell identity $\tau=(L1+L2)/2$ both lattices